Unit 10, Lesson 6: Solving Problems Involving Inscribed Angles

Benchmark

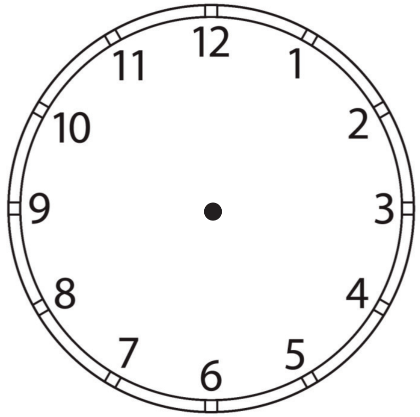
MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.

Learning Target

☐ I can use the relationship between the measure of an inscribed angle and its intercepted arc to solve problems.

10.6.1 Warm-Up: Angles on a Clock

1. Use the clock to answer the following questions.



a. If the time is $10{:}10$ and the hour hand does not move, what is the exact angle between the minute hand and hour hand?

b. If the hour hand does move, what is the exact angle if the time is $7{:}30$?

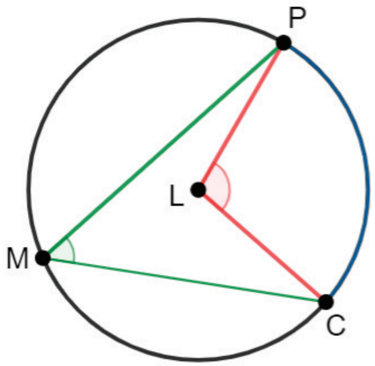
c. If the hour hand does move, what is the exact angle if the time is $7{:}35$?

10.6.2 Exploration: Inscribed Angles

1. Circle L is shown where P, C, and M are points on the circle.

A ***chord*** is a line segment on the interior of a circle with both endpoints lying on the circle.

An ***inscribed angle in a circle*** is an angle which is formed in the interior of a circle when two chords share an endpoint.



Central Angle $m\angle PLC = 110°$

Inscribed Angle $m\angle PMC = 55°$

Arc Measure $\widehat{PC} = 110°$

a. What is the connection between the central angle and the arc measure?

b. Describe the relationship between the central angle and the inscribed angle.

c. Write a conclusion about the measures of the central angle, the inscribed angle, and the arc measure.

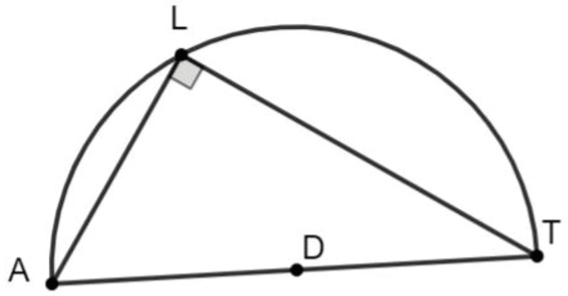
d. Share your response to Part C with two classmates. Listen to their conclusion and summarize it in the table. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating your summary was correct.

My Summary of Their Conclusion	Initials

e. Using Circle L, complete the table.

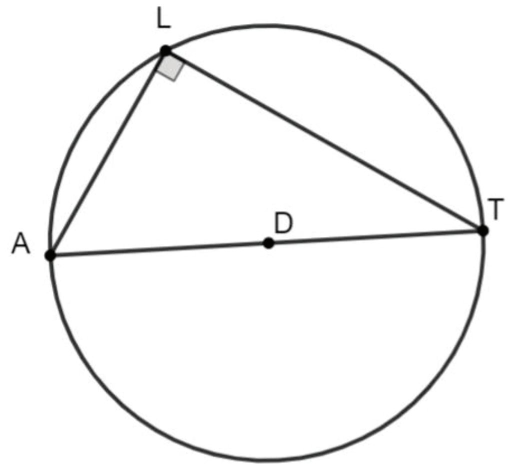
Central Angle	Inscribed Angle	Arc Measure
$216°$		
	$81°$	
		$143°$
	$145°$	

2. $\angle ALT$ is inscribed in semicircle D, as shown.



a. Describe the angle inscribed in the semicircle.

b. If the entire Circle D is shown, what is the measure of $\widehat{AT}$?

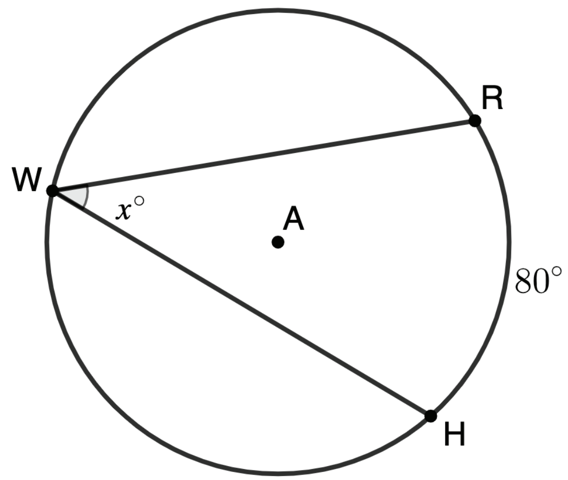


c. Discuss with your partner the relationship between an inscribed angle and a semicircle.

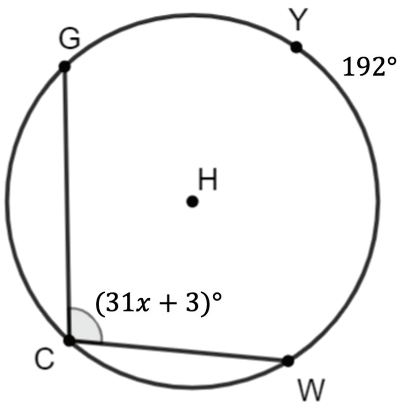
Inscribed Angle Theorem
The measure of an inscribed angle is one-half the measure of the intercepted arc.

10.6.3 Guided Practice: Inscribed Angles

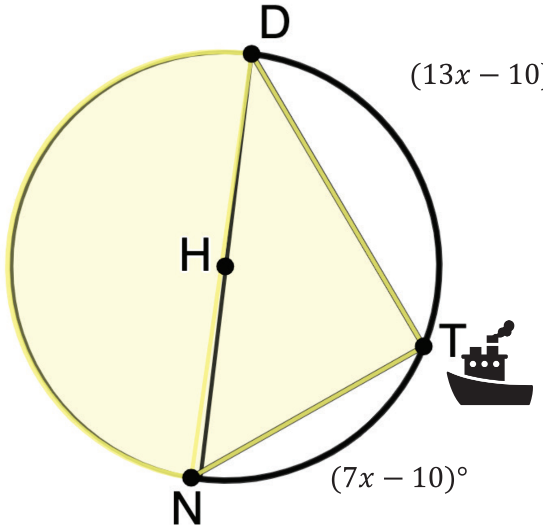
1. Circle A is shown where points R, W, and H are on the circle. $\angle RWH$ is inscribed inside Circle A with $m\,\widehat{RH} = 80°$. Find the value of x.



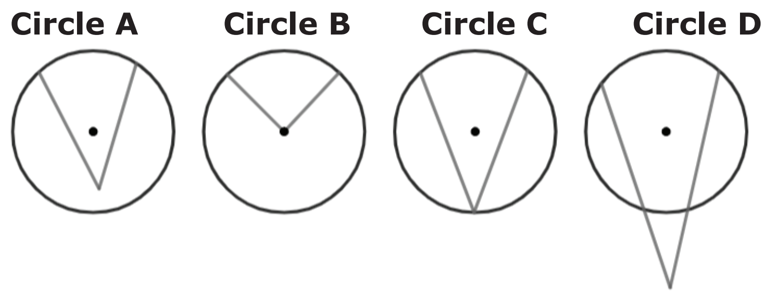
2. Points Y, G, C, and W lie on Circle H. $\angle GCW$ is inscribed in the circle. Find x, given $m\,\widehat{GYW} = 192°$ and $m\angle GCW = (31x + 3)°$.



3. Boats use angle measures and arcs when sailing in treacherous waters. When a boat captain turns on the boat light, it forms inscribed angle $\angle NTD$. Use the information for Circle H to find $m\angle NDT$.



4. Craig and Valentina are reviewing circles to identify inscribed angles.

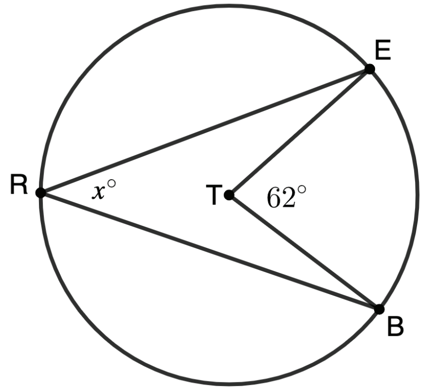


Valentina identified Circle C is an inscribed angle. Craig identified both Circle A and Circle C as inscribed angles.

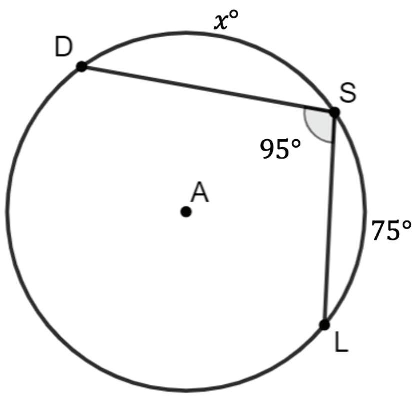
Do you agree with Valentina, Craig, or neither? Justify your reasoning.

10.6.4 Your Turn!

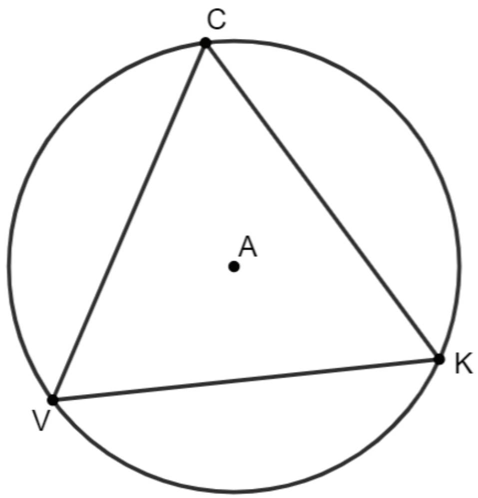
1. Circle T is shown where $\angle ETB$ is a central angle, $\angle ERB$ is inscribed in the circle, and $m\angle ETB = 62°$. Find $m\angle ERB$.



2. Circle A has points D, S, and L on the circle. $\angle DSL$ is inscribed in Circle A measuring $95°$. Given $m\widehat{SL} = 75°$, find $m\widehat{DS}$.

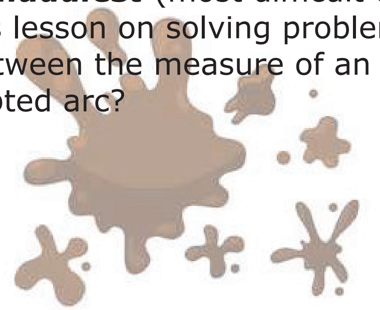


3. Equilateral triangle CKV is inscribed in Circle A. Circle A consists of three equal inscribed angles. Determine the value of the inscribed angle.



10.6.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on solving problems using the relationship between the measure of an inscribed angle and its intercepted arc?



Unit 10, Lesson 7: Solving Problems Involving Intersecting Chords

Benchmark

MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.

Learning Target

☐ I can solve problems involving chords of a circle and their related angles.